Healing
شفاء

Doctor of Pharmacy 2016

Calculus 102A

First exam

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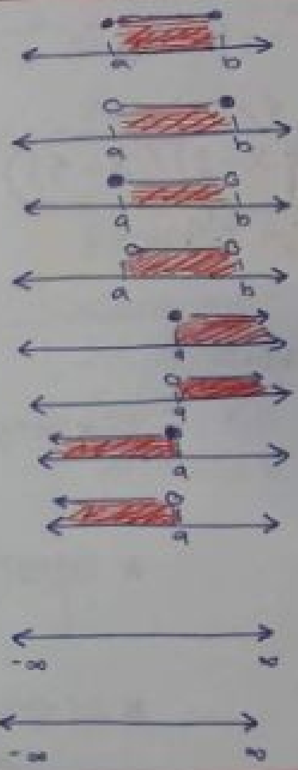
#نبض_من_الشفاء_ارتوى

الموضوع الأول ~ الفترات

① Intervals

Intervals	Description	Graph	Example
أقفل $[a, b]$	$a \leq x \leq b$		$[1, 2]$
$(a, b]$	$a < x \leq b$		$(1, 2]$
$[a, b)$	$a \leq x < b$		$[1, 2)$
(a, b)	$a < x < b$		$(1, 2)$
$[a, \infty)$	$x \geq a$		
(a, ∞)	$x > a$		
$(-\infty, a]$	$x \leq a$		
$(-\infty, a)$	$x < a$		
$(-\infty, \infty)$	R الأعداد الحقيقية	$-\infty$ ∞	
$\{\ \}$	$\emptyset$	$-\infty$ ∞	



[2] Solving Liner equality (=) and in-equality (≠)

steps solution

1- The right side is equal to zero.

2- find the critical numbers. → البسط = : / المقام = :

3- study the sign of ...

② Solving liner equality and in-equality.

Ex: solving the following inequalites:

① $4x + 5 < 6x - 9$

② $x^2 - 4 \geq 0$

③ $\dfrac{(x-1)}{(x-3)(x-5)} \geq 0$

④ $\dfrac{1}{x} \leq 1$

آ ركز كثير ☺

Solution: ①

* step (1): $4x + 5 < 6x - 9$

$-2x + 14 < 0$ → ندرس المنطقة السالبة من المجال

* step (2): $-2x + 14 = 0$

$x = 7$

* step (3): sign of $-2x+14$:

$+ \quad 7 \quad -$

Solution $(7, \infty)$

② * step (1): ⇒ $x^2 - 4 \geq 0$ → ندرس المنطقة الموجبة من المجال.

* step (2): $x^2 - 4 = 0 \Longrightarrow x = \pm 2$

* step (3): sign of $x^2 - 4$:

$+ \quad -2 \quad - \quad 2 \quad +$

solution : $(-\infty, -2] \cup [2, \infty)$.

③ * step (1): [illegible]

* step (2): Expression $= 0 \Rightarrow x - 1 = 0 \Rightarrow x = 1$

مقام place $= 0 \Rightarrow x = 3 \ , \ x = 5$

لا نضع مساواة لأنها أصفار مقام

* step (3): sign of $\dfrac{(x-1)}{(x-3)(x-5)}$

$- \quad 1 \quad + \quad 3 \quad - \quad 5 \quad +$

solution $[1, 3) \cup (5, \infty)$.

④ * step (1): $\frac{1}{x} - 1 \leq 0 \Rightarrow$ ندرس المنطقة السالبة من المجال

* step (2): $\frac{1}{x} - 1 = 0 \Rightarrow x = 1 , x = 0$ بعد توحيد المقامات

* step (3): $x \neq 0$ sign of $\frac{1-x}{x}$: $- - - \overset{\circ}{0}\ + + +\ \overset{\bullet}{1}\ - - -$

solution : $(-\infty, 0) \cup [1, \infty)$.

Remark ...

معادلة > 0 المجال الموجب

معادلة ≥ 0 المجال الموجب والصفر

معادلة < 0 المجال السالب

معادلة ≤ 0 " " والصفر

$\mathbb{R}$ $\mathbb{Q}$

Ex: solv for x.

* ① $1 + x^2 > 0$
* ② $1 + x^2 < 0$

③ $\frac{(x-2)^2}{(x-3)} \geq 0$

* ④ $2 - 3x \leq 2x + 7 < 3x - 1$

Solution

1 step(1): – $\Rightarrow$ $1 + x^2 > 0$

step(2): $1 + x^2 = 0 \Rightarrow x^2 \neq -1$ $\notin \mathbb{R}$

step(3): $-\infty\ + + + + +\ \infty$

solution : $\mathbb{R}$

2 مماثلة لـ ① لكن لا يوجد
قيم للمجال (+). القيم المراد
دا سالبة (−).

شونكتب الجواب ؟

3 $(x-2)^2 = 0 \Rightarrow x = 2$

$(x-3) = 0 \Rightarrow x = 3$

$-\ \overset{\bullet}{2}\ -\ \overset{\circ}{3}\ +$

④ $2 - 3x \leq 2x + 7 < 3x - 1$

Intersection

تقاطع الإشارات المشتركة

a: معادلة
b: معادلة
c: معادلة

Remark

1- $a \leq b \leq c$: $a \leq b \cap b \leq c$

تلزمنا هنا → $a \leq b < c$: $a \leq b \cap b < c$

$a < b \leq c$: $a < b \cap b \leq c$

$a < b < c$: $a < b \cap b < c$

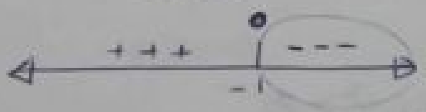
$2 - 3x \leq 2x + 7 \quad \cap \quad 2x + 7 < 3x - 1$

نجد مجالها لوحدها | نجد مجالها لوحدها

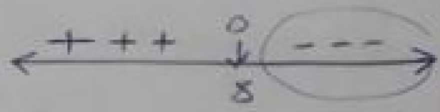
$-5 - 5x \leq 0$ | $-x + 8 < 0$

$x = -1$ | $x = 8$

Sign of $-5 - 5x$



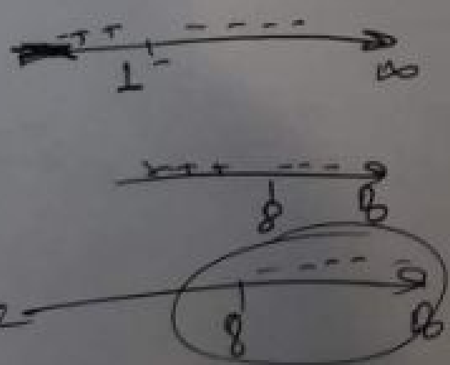
$\cap$



solution : $[-1, \infty)$ | solution $(8, \infty)$.

$\therefore$ solution $(8, \infty)$.

لاحظ: المنطقة السالبة المشتركة



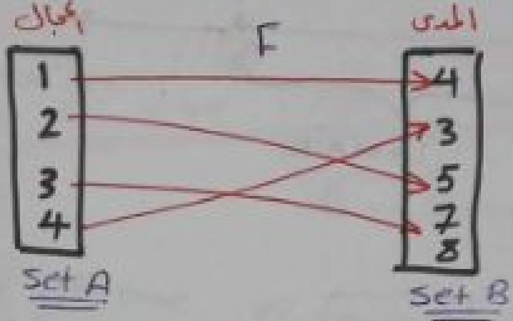
③ Functions الاقترانات

الاقترانات

y = f(x)
(ص = ق(س))

* يطلب منك ايجاد Domain ← و Range لكل اقتران
(Domain: المجال، Range: المدى)

* الاقتران: لكل عنصر في المجال صورة واحدة في المدى.



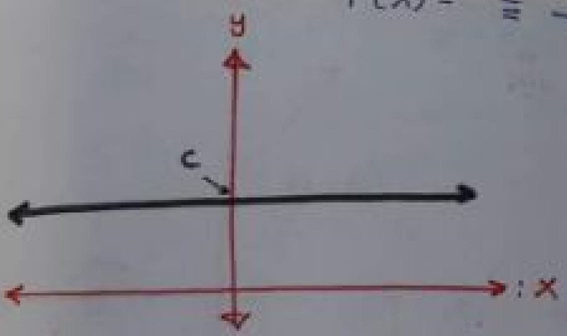
* Set A is called domain (المجال) of (F)
Set B is called co-domain (المجال المقابل) of (F)
{3, 4, 5} is called Range (المدى) of (F)

نفس بعض
range ⊆ codomain

Examples of Common functions

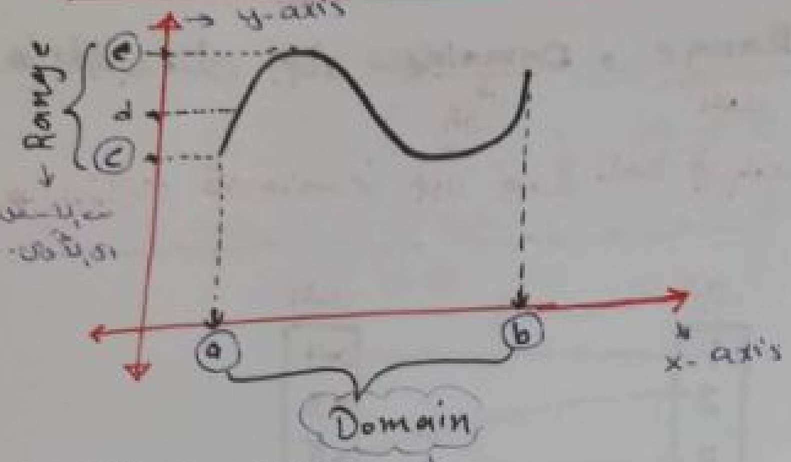
Ⓐ Constans function : الاقتران الثابت

f(x) = c → Constan st (الثابت)



* Domain : R.
Range: {c}

* Domain and Range graphicall «بيانياً»



⇒ $\boxed{\text{Domain}}$ is starting pt (point) → ending of.

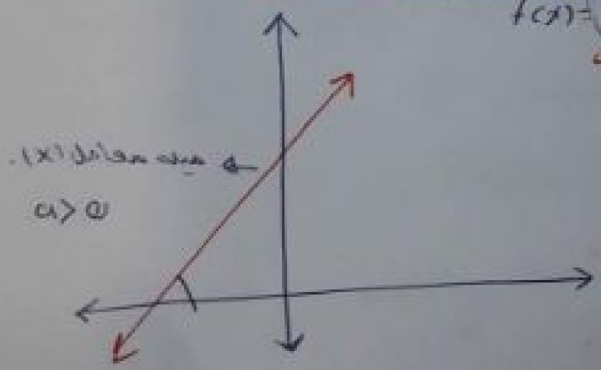
⇒ $\boxed{\text{Range}}$ min pt (point) to → max

EX: $f(x) = 3$
Domain $= R$
Range $= \{3\}$.

EX: $f(x) = -1$
Domain $= R$
Range $= \{-1\}$.

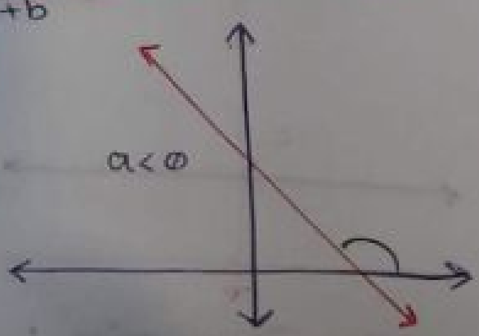
Liner function... الاقتران الخطي

$f(x) = ax + b$
الميل



Domain = R

Range = R



Domain R.

Range R.

EX: Find the domain and Range of :

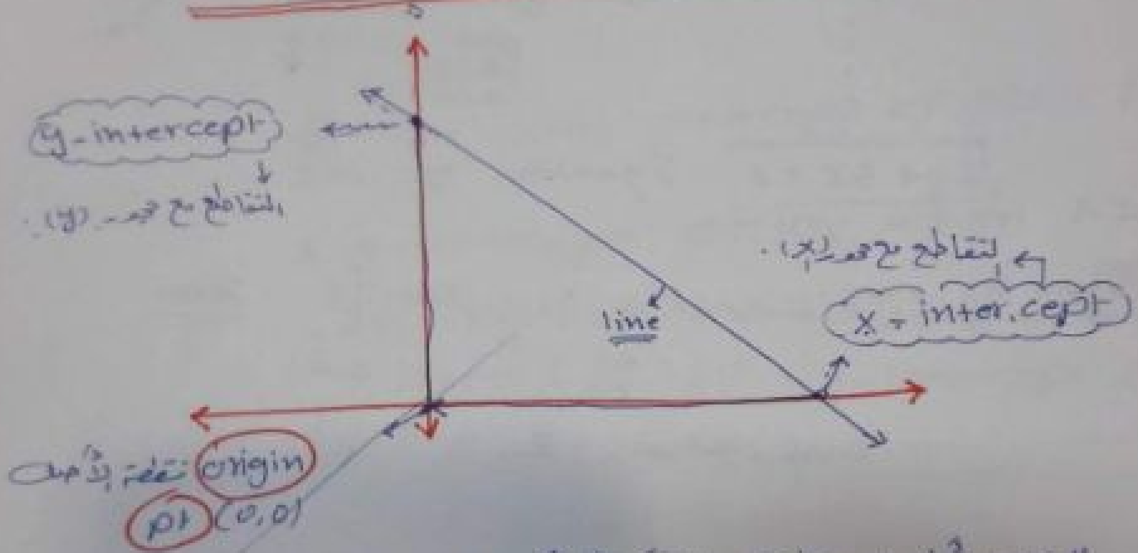
① $f(x) = 4 - 3x \longrightarrow a = -3 \ , \ b = 4$

② $f(x) = 5x \longrightarrow a = 5 \ , \ b = 0$

Solution ① + ② Domain of $f(x) = \mathbb{R}$

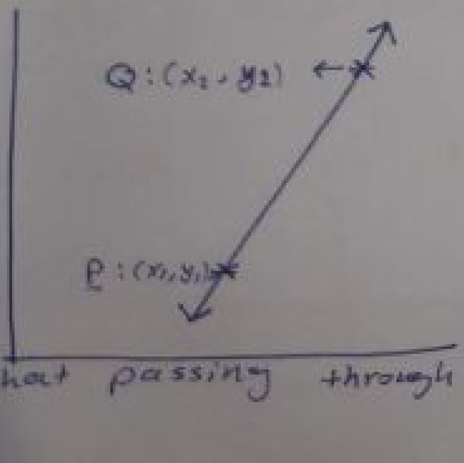
Range of $f(x) = \mathbb{R}$

Equation of the liner function



* يجب أن يمر بنقطتين داخ ... باستثناء اذا مر بنقطة الأصل

* اذا مر بنقطة الأصل ومش معلوم ميله ... نستخدم أي ~~نقطتين~~ نقطة أخرى.



* slope of the line that passing through $p(x_1, y_1)$ and $Q : (x_2 - y_2)$:

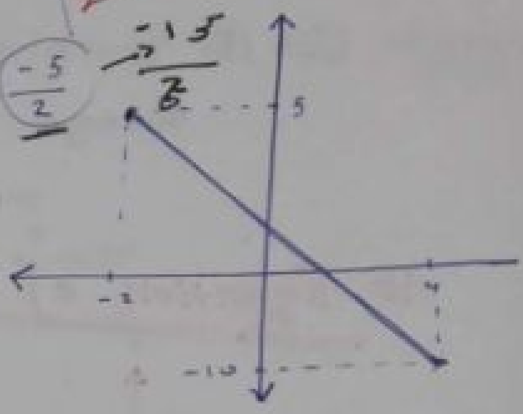
$$m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1} = \text{slope.}$$

* Equation of the liner that passing through $p(x_1, y_1)$ and $Q : (x_2, y_2)$

معادلة الخط المستقيم $\boxed{(y - y_1) = m(x - x_1)}$

EX: Find the equation of the liner that passing throught $p(\overset{x_1}{-2}, \overset{y_1}{5})$ and $q(\overset{x_2}{4}, \overset{y_2}{-10})$

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-10-5}{4--2} = \frac{-5}{2} \;\; \frac{-15}{6}$$



Equation : $(y - y_1) = m(x - x_1)$

$$(y-5) = \frac{-5}{2}(x+2)$$

$$\boxed{2y + 5x = 0} \text{ Equation of line.}$$

ملاحظة: اذا اعطاك السؤال ان الخط يمر بنقطتين نستفيد من ذلك

↲ ايجاد الميل $\left(\frac{\Delta y}{\Delta x}\right)$.

↲ (x_1, y_1) : نعوضها في المعادلة .

y-intercept: →   ملاحظة : نضع $x = 0$

x-intercept: →   نضع $y = 0$

EX: Find the x-intercept and y-intercept of:

① $2x + 3y = 18$.

② $\frac{x}{5} + \frac{y}{3} = 10$.

<u>Solution</u>:

① x-intercept occure (تحدث) when $y = 0$.

→

→ $2x + 3y = 18 \rightarrow 2x = 18$

$\boxed{x = 9}$

∴ the point intercept of x axis {9, 0}.

* y-intercept occure when $x = 0$

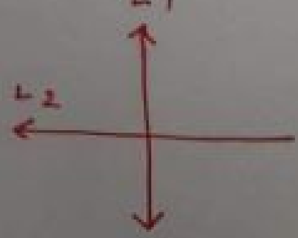
$2x + 3y = 18 \rightarrow 3y = 18 \rightarrow \boxed{y = 6}$

The point of y-intercept (0, 4).

(parallel)): التوازي

$L_1 \mathbin{/\!/} L_2$ (يوازي) → slope of L_1 = slope of L_2

(perpendicular): التعامد



$L_1 \perp L_2$
→ slope of L_1 × slope of L_2 = −1

EX: Find the Equation of the line has

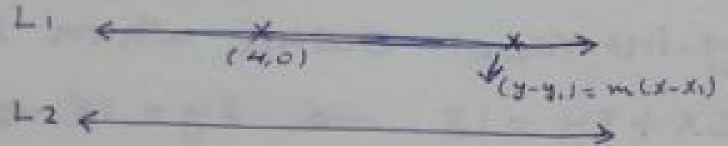
(4, 0) ① x-intercept 4 and parallel to line [(1 - 3x = y)] ? (4, 0)

② x-intercept −3 and y-intercept 1/2.
(−3, 0) (0, 1/2)

③ passes (−3, 1) and perpendicular $(1 - \frac{x}{2} = y)$.

solution: Remark: The slope of ($y = mx + b$) is m

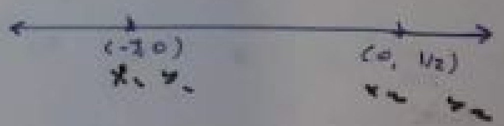
① slope of L_1 = slope of L_2 = −3



So the equation of L_1 is:

$$(y - y_1) = m(x - x_1)$$
$$(y - 0) = -3(x - 4).$$
$$y = -3(x-4) \rightarrow \boxed{y = -3x + 12}$$

②



$$\text{slope} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\frac{1}{2} - 0}{0 - -3}$$
$$m = \frac{1}{6}$$

So the equation

$$(y - y_1) = m(x - y_1)$$
$$y = \frac{1}{6}(x - -3)$$

③ slop of $L_2 = \frac{-1}{2}$

slope of L_1 * slope of L_2 = −1

solpe of $L_1 \times \frac{-1}{2} = -1 \Rightarrow \boxed{\text{slop } L_1 = 2}$

Equation → $y - 1 = 2(x - -3)$
$y = 2x + 5$

Ex: لنكن **Let** $2(1-3x)+y=0$ **Find:-**

$u - x + 5$

x	zero	-5

① slope of the line

② x-intercept.

③ y-intercept.

④ ارسم sketch of the line.

solution:

① $2-6x+y=0 \rightarrow y = 6x-2 \Rightarrow$ slope $= 6$

حولناها بالصورة العامة لكي نستطيع استخراجه.

يمكن من البداية نستطيع استخراجه

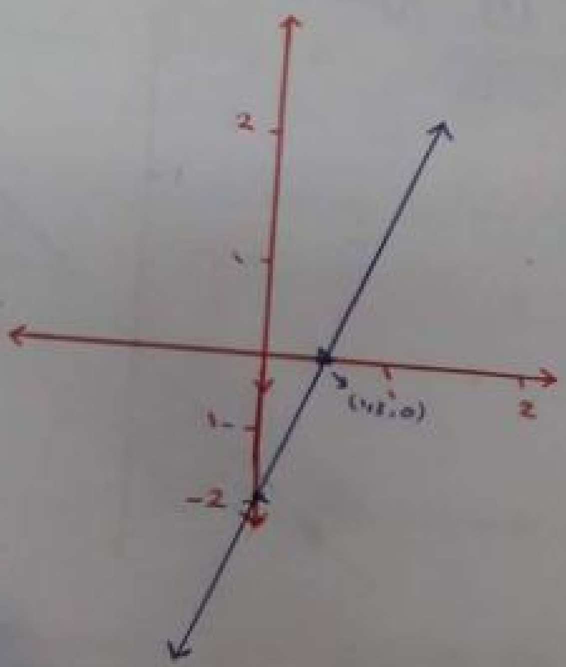
$-\frac{\text{(x) معامل}}{\text{(y) معامل}}$.

② x-intercept when $y=0 \rightarrow 6x=2$

$x = 1/3 \rightarrow (1/3, 0)$

③ y-inter-cept when $x=0 \Rightarrow y=-2 \Rightarrow (0,-2)$

④



* solving system of liner inqualites graphically.

EX: sketch the solution of the following

Ⓐ $\boxed{y \le x}$

Ⓑ $y \le x + 5$, $y > 4 - 2x$

Ⓒ $x + y > 1$, $4 - 3y \le 0$, $2x + 3y < 6$

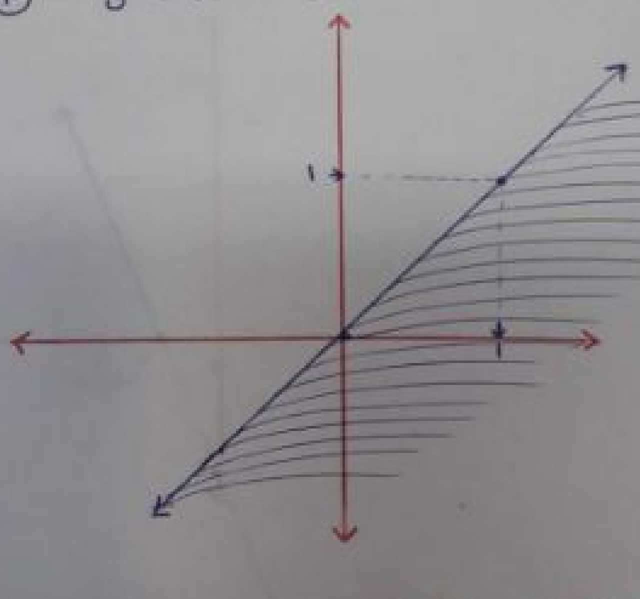
Ⓓ $x \le 4$, $y > 2$, $x + y > 6$

Solution:

① $y \le x$; $y = x$ $\longrightarrow$

x	zero	zero	1
y	zero	zero	1

* للأسف أخذت نفس النقطة ولا تكون خط مستقيم



$\longleftarrow$ المنطقة التي حققت المتباينة

لمعرفة أي المساحتين هي مساحة الحل:

↲ نعوض نقطة من المساحة السفلى في المتباينة، اذا حققت المتباينة تكون المنطقة السفلي هي منطقة الحل ... واذا لم تحقق .. مباشرة تكون المنطقة العليا هي منطقة الحل.

أسهل نقطة للتعويض هي (0،0) .. لكن لسوء الحظ في هذا المثال هي نقطة فصل.

Solution (2): $\rightarrow\ y \le x+5$; $y = x+5$

x	zero	-5
y	5	zero

(0.5) (-5, 0)

$\rightarrow\ y > 4-2x$, $y = 4-2x$

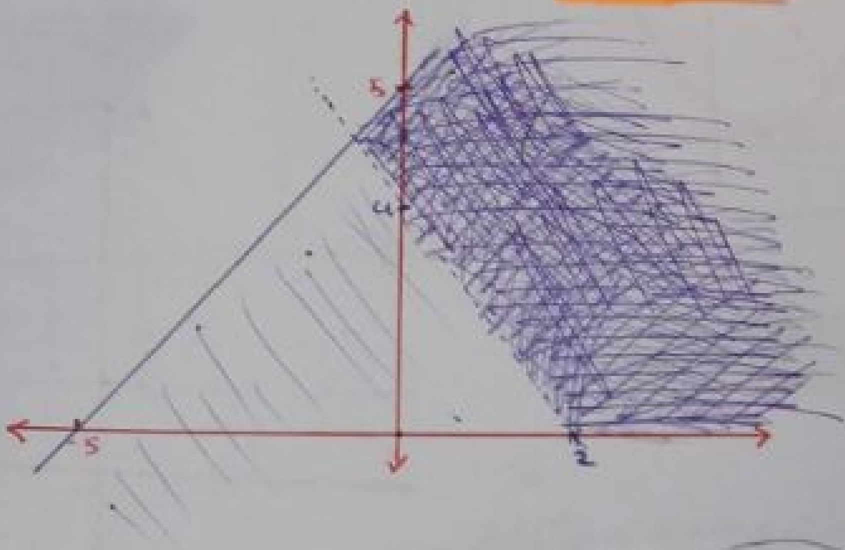
x	zero	2
y	4	zero

(0. 4) (2, 0)

ملحظ :

$\le$: يكون خط غير متقطع.

$<$: يكون خط متقطع.



Solution (3): $\rightarrow\ x+y > 1$; $x+y=1$

x	zero	1
y	1	zero

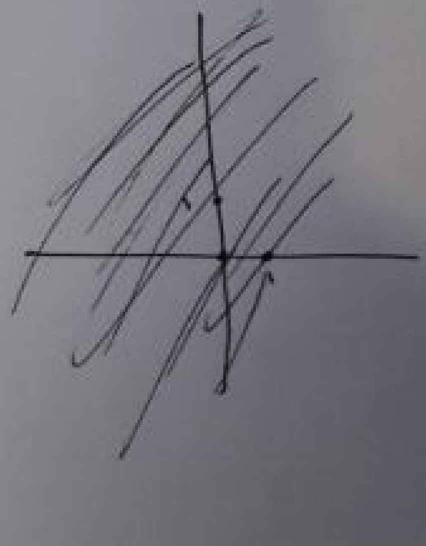
$\rightarrow\ 4-3y \le 0$; $4-3y=0$

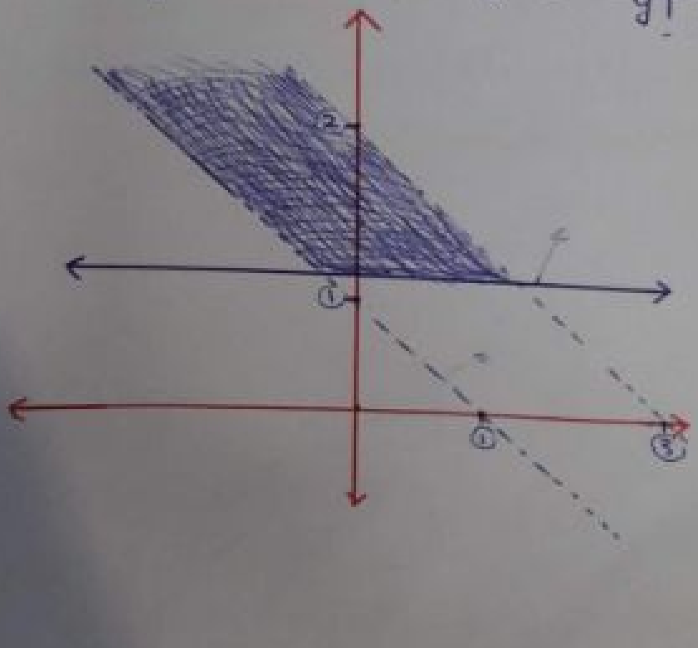
$3y = 4 \rightarrow y = \frac{4}{3}$

x	zero	
y		zero

$\rightarrow\ 2x+3y < 6 \rightarrow 2x+3y=6$

x	zero	3
y	2	zero





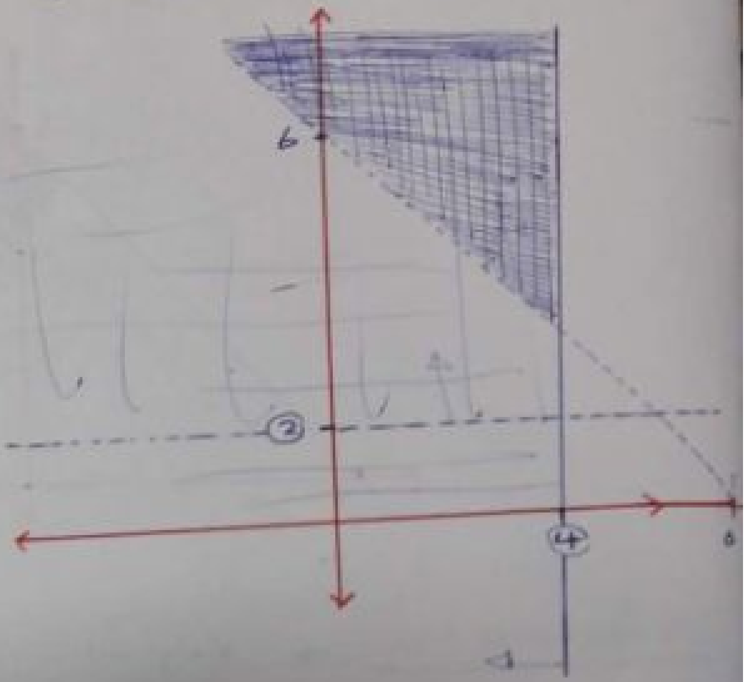
Solution (4):

$\rightarrow x \leq 4 \longrightarrow x = 4 \leftarrow$ خط متصل

$\rightarrow y > 2 \longrightarrow y = 2 \leftarrow$ خط متقطع

$\rightarrow x + y \geq 6 \Rightarrow x + y = 6$

x	zero	6
y	6	zero



(14)

3 Quadratic function

parabola القطع المكافئ

$f(x) = ax^2+bx+c$ الاقتران التربيعي

$\left(\frac{-b}{2a}, f\left(\frac{-b}{2a}\right)\right)$

→ Domain of $f(x)$ is $\mathbb{R}$

→ Range of $f(x)$: → $\boxed{a>0}$ ⇒ Range of $f(x) = \left[f\left(\frac{-b}{2a}\right), \infty\right)$

→ $\boxed{a<0}$ ⇒ Range $f(x) = \left(-\infty, f\left(\frac{-b}{2a}\right)\right]$

EX: let $f(x) = x^2 - 9$ Find :

① Domain and Range of $f(x)$.

② vertex

③ x-intercept AND y-intercept

④ sketch of $f(x)$.

① Domain of $f(x) = \mathbb{R}$

Range of $f(x) = \left[f\left(\frac{-b}{2a}\right), \infty\right)$

$\left[f\left(\frac{0}{2}\right), \infty\right)$

$[-9, \infty)$

② vertex $\left(\frac{-b}{2a}, f\left(\frac{-b}{2a}\right)\right) = (0, -9)$

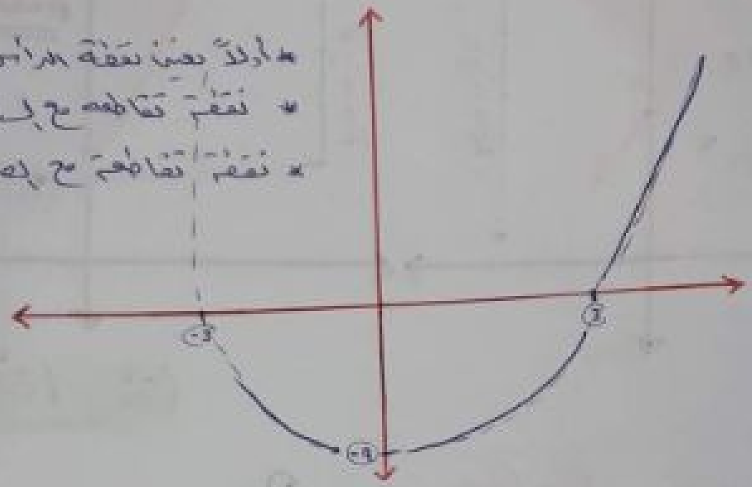
③ x-intercept → ((when $y=0$)) → $x=3$, $x=-3$

y-intercept ((" $x=0$)) $y=-9$

④ * أولاً نعين نقطة الرأس

* نقط تقاطعه مع السينات

* نقط تقاطعه مع الصادات.



HW: Repeat the previous Example for

① $f(x) = 4x - x^2$. → $a=-1$, $b=4$

x-intercept ← ② $f(x) = x^2 + 9$. → $a = 1$ $b = 0$

solution ①:

① Domain $= \mathbb{R}$

Range $= (\infty, f(\frac{-b}{2a})] = (\infty, f(\frac{4}{2})$

$(\infty, f(2)] = (\infty, 4]$

② vertex: $\left(\frac{-b}{2a}, f\left(\frac{-b}{2a}\right)\right) = (2, 4)$.

③ x-intercept: $x(4-x) = 0 \rightarrow x=0$, $x=4$

y-intercept: $f(x) = 0 \rightarrow y=0$

Sets: المجموعات

* **IN** : Natural number الأعداد الطبيعية
لماذا كنت بين أصابعك طبيعية

$\mathbb{N} = \{1, 2, 3, \dots\}$.

فقط الموجبة

* **I/Z**: Integer numbers الأعداد الصحيحة

$I = \{-1, -2, -3, \dots, 0, +1, +2, +3 \dots\}$

(موجب وسالب)

* **Rational numbers**: $\mathbb{Q}$

$\mathbb{Q} = \left\{ \frac{a}{b} : a \text{ and } b \neq 0 \text{ are the both integ} \right.$

for Example: $\frac{-2}{3}$, $\frac{1}{2}$, $4 \to \frac{4}{1}$, $0.2 \to \frac{2}{10}$, $0.\overline{3}$

* $\mathbb{R} = (-\infty, \infty)$ Real numbers الأعداد الحقيقية

$\mathbb{N} \subseteq I \subseteq \mathbb{Q} \subseteq \mathbb{R}$

$\in$: belongs to / in / ينتمي

$\subseteq$: sub set احتواء

$4 \in \mathbb{R}$ ✓

$3/4 \in \mathbb{N}$ ✗

$\{1\} \subseteq \{1, 2, 3\}$ ✗

$\{\}$ لازم تكون داخل

$\{1\} \subseteq \{1, 2, 3\}$. ✓

$\subseteq$

بين مجموعتين

Ex: Use The listing method to describe The following sets:

① The sets of all integers less than 5 and greater than -2.

$\{-1, 0, 1, 2, 3, 4\}$.

② The sets of all natural number less than 50.

$\{1, 2, 3, \dots, 49\}$.

هي كل الاعداد التي تقبل القسمة على نفسها، والواحد فقط

عدا الواحد

③ the set of all prime numbers less than 20.

الأعداد الأولية

$\{2, 3, 5, 7, 11, 13, 17, 19\}$.

④ The set of all even numbers less than 100.

زوجية

$\{\dots, -4, -2, 0, 2, 4, 6, \dots, 98\}$.

ملاحظة

even numbers

$\{\dots, -4, -2, 0, 2, \dots\}$

odd numbers

$\{\dots, -5, -3, -1, 1, 3, 5\}$.

positive even numbers.

$\{2, 4, 6, 8, \dots\}$.

EX: which of the following statement (الجمل - العبارات) are true and wich are false.

① $2 \in \{1, 2, 3\}$. True
② $3 \subseteq \{1,2,3\}$. false
③ $\{3\} \subseteq \{1,2,3\}$. True
④ $\{5,6\} \subseteq \{1, 2, 5, 6\}$. True

---

④ **Rational function** ⟸ الاقترانات النسبي

↳ فقط بندرس: البسط كثير حدود. والمقام كثير حدود. ~~خطي على خطي~~.

$$f(x) = \frac{ax+b}{cx+d}$$

$$f(x) = \frac{p(x)}{q(x)}$$

where $p(x)$ and $q(x) \neq 0$ are poth-poly numbers.

← سنتعرف في د. إرتنا طرح المادة فقط على الاقترانات النسبية اللي يكون فيها البسط والمقام اقترانات خطية.

* Domain : $R - \left\{\frac{-d}{c}\right\}$ صفر المقام من مستحيلات الاقتران

* Range : $R - \left\{\frac{\text{cofficient of } (x) \text{ في البسط}}{\text{cofficient of } (x) \text{ في المقام}}\right\} = \left\{\frac{a}{c}\right\}$

EX: Determine the domain and range of

① $f(x) = \boxed{\frac{2x+4}{3x-6}}$ → Domain ⇒ $R - \{2\}$.
→ Range ⇒ $R - \{\frac{2}{3}\}$.

② $f(x) = \frac{4x}{x-2}$ → $R - \{2\}$.
→ Range $R - \{4\}$.

③ $f(x) = \frac{2}{(x+1)}$ → Domain ⇒ $R - \{-1\}$
→ Range ⇒ $R - \{0\}$.

5

n^{th} root function

إقتران الجذر النوني

$$\boxed{f(x) = \sqrt[n]{g(x)}}$$

condition

→ Domain = $\{x : g(x) \geq 0$ if n = even $\}$.

~~Range =~~

→ Domain = { Domain $g(x)$ if n = odd }

EX: Find the domain of following:

① $f(x) = \sqrt[2]{x^2 - 4}$

② $f(x) = \sqrt{\dfrac{x-1}{x+4}} \geq 0$

H.W ③ $f(x) = \sqrt[4]{\dfrac{x-1}{(x-3)(x-5)}} \geq 0$

H.W ④ $f(x) = \sqrt[3]{\dfrac{(x-1)}{(x-4)}}$

Solution ①: $\sqrt{x^2 - 4}$; $x^2 - 4 \geq 0$

$x^2 - 4 = 0 \rightarrow x = 2,\ x = -2$

Sign of: $x^2 - 4$

$(-\infty, -2)$	-2	$(-2, 2)$	2	$(2, \infty)$
+		−		+

Domain: $(-\infty, -2] \cup [2, \infty)$

Solution ② $\sqrt{\dfrac{(x-1)}{(x+4)}}$; $\dfrac{(x-1)}{(x+4)} \geq 0$

$\dfrac{(x-1)}{(x+4)} = 0 \rightarrow x = 1,\ x = -4$

Sign of $\dfrac{(x-1)}{(x+4)}$

(4) صفر المقام

$(-\infty, -4)$	-4	$(-4, 1)$	1	$(1, \infty)$
+		−		+

Domain = $(-\infty, -4) \cup [1, \infty)$.

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Solution ③ : $f(x)=\sqrt[4]{\frac{(x-1)}{(x-3)(x-5)}}$; $\frac{(x-1)}{(x-3)(x-5)} \geq 0$

$\frac{(x-1)}{(x-3)(x-5)} = 0 \rightarrow x=1$, $x=3$, $x=5$

مرفوضة

Sign of $\frac{(x-1)}{(x-3)(x-5)}$: $\quad - \;\; \underset{1}{|} \;\; + \;\; \underset{3}{|} \;\; - \;\; \underset{5}{|} \;\; +$

Domain $[1, 3) \cup (5, \infty)$.

Solution ④ : $f(x)=\sqrt[3]{\frac{(x-1)}{(x-4)}}$; = Domain of $\left(\frac{x-1}{x-4}\right)$

=

Applications
Quadratic function.

عدفي

EX: Find the numbers (x) and (y) that satisfy the condition that $(3x-y=18)$ and for wich their product xy is as small as possible.

ضرب

Solution: $P = xy$; but $3x-y=18 \rightarrow y = 3x-18$

$P = x(3x-18) \rightarrow P(x) = 3x^2-18x$

$\boxed{\text{or } P(x)=3x^2-18x}$ quadratic function. with $a=3$, $b=-18$

ملاحظة : نستطيع اخراج قيمة (x) عن طريق الاشتقاق

vertex $= \left(\frac{-b}{2a}, P\left(\frac{-b}{2a}\right)\right) = \left(\frac{18}{6}, P\left(\frac{18}{3}\right)\right)$

$= (3, P(3)) = (\underset{x}{3}, \underset{P}{-27})$

x قيمة

قيمة P

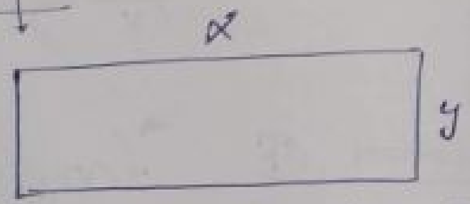
or نخرج قيمة y عن طريق أي معادلة

But; $\Rightarrow 3x-18 = y$

$y = 3(3)-18$

$\boxed{y=-9}$

مزارع
EX: Afarmar has 400 yd of fencing with which to bulid arectanglar field. حقل مستطيل الشكل
what is the largest area he can enclose?



solution area $= xy$

$A = xy$

$A = xy$; but $2x + 2y = 400 \Rightarrow x + y = 200$

$A = x(200 - x)$; $A = 200x - x^2$

← شكل البيانات معامل x سالب
الى الاسفل.

$\boxed{A = 200x - x^2}$ Quadratic function with $a = -1$, $b = 200$

vertex $= \left(\frac{-b}{2a}, A\left(\frac{-b}{2a}\right)\right) = (100, A(100))$

$= (\underset{x}{100}, \underset{A}{10000})$

قيمة x ← قيمة A

But: $y = 200 - x$

$y = 200 - 100 = 100$

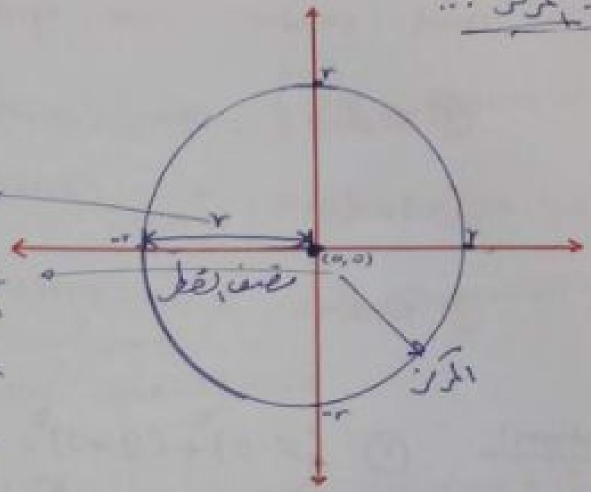
(largest)

ملاحظة: عادة السؤالات السابقة كانت فإن معامل x^2 موجب.

6

Circle الدائرة

١- نصف القطر ...
٢- المركز ...



نصف القطر ... radius

center
المركز

Equation: $x^2 + y^2 = r^2$

معادلة الدائرة التي مركزها نقطة الأصل

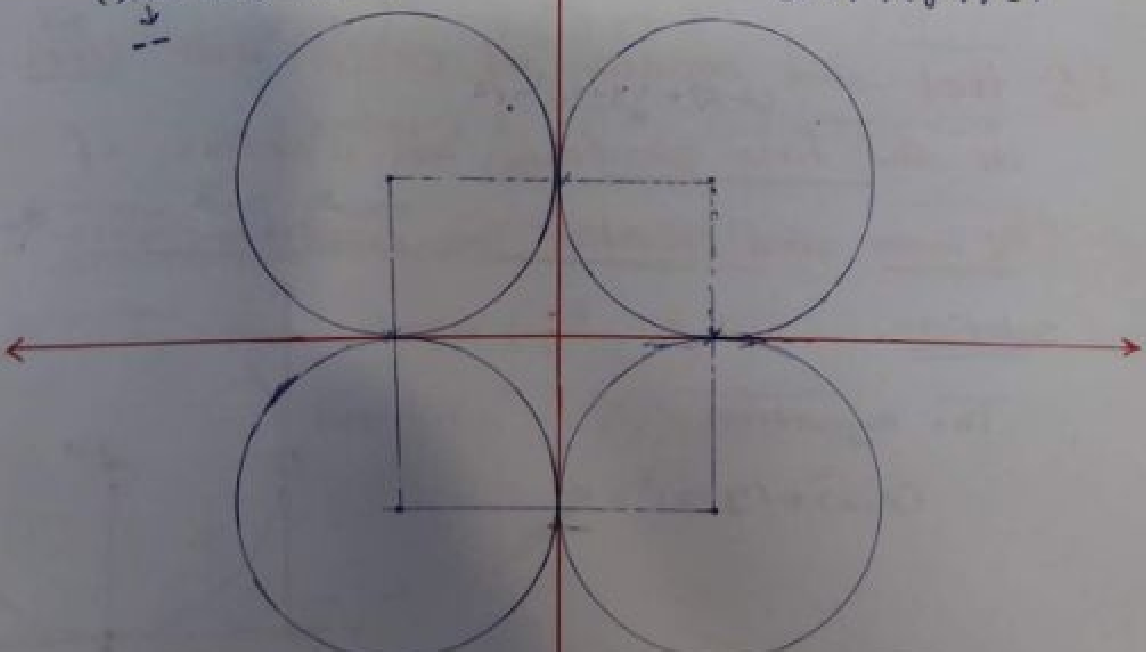
$$(x-r)^2 + (y-r)^2 = r^2$$

2nd Quadratic الربع الثاني

$(x+r)^2 + (y-r)^2 = r^2$

First Quadratic الربع الأول

$(x-r)^2 + (y-r)^2 = r^2$



3th Quadratic الربع الثالث

$(x+r)^2 + (y+r)^2 = r^2$

4th Quadratic الربع الرابع

$(x-r)^2 + (y+r)^2 = r^2$

EX: Find the quation of circle whose radii (أنصاف أقطار) and center are given.

① radius = 5, Center = $(0, 2)$ (x_1, y_1)

② radius = 4, Center = $(-3, 0)$ (x_1, y_1)

③ radius = 2, Center = $(0, 0)$ (x_1, y_1)

Solution:

① $(x-0)^2 + (y+2)^2 = (5)^2$

$x^2 + (y+2)^2 = 25$

② $(x+3)^2 + y^2 = 16$

③ $x^2 + y^2 = 4$

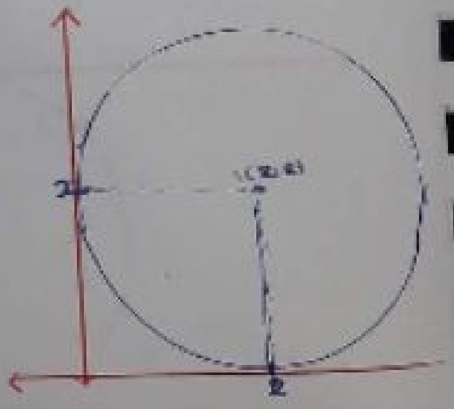
EX: find the equation of circle that lines (تقع) in the $\boxed{\text{first quadratic}}$ has a radius of 2 units and touches the coordinate axis (تمس المحورين).

$(x-x_1)^2 + (y-y_1)^2 = r^2$

Solution

The equation:

$(x-2)^2 + (y-2)^2 = 4$



24

EX: Find the center and radius of the followi[ng] circls :

① $y^2+(x-3)^2=7$; ⇒ Center $(3, 0)$, radius $=\sqrt{7}$

② $x^2+y^2-4=0$ → Center $(0,0)$, radius $=2$

③ $x^2+y^2-4x+6y-11=0$ اكمال مربع

by completing the square.

$$\left[x^2-4x+\left(\frac{-4}{2}\right)^2-\left(\frac{-4}{2}\right)^2\right]+\left[y^2+6y+\left(\frac{+6}{2}\right)^2-\left(\frac{+6}{2}\right)^2\right]$$

$$(x^2-4x+4-4)+(y^2+6y+9-9)=11$$

$$(x-2)^2+(y+3)^2=11+4+9$$

$$(x-2)^2+(y+3)^2=24$$

Center $(2,-3)$, radius $=\sqrt{24}$

Remark: $x^2+ax+y^2+by+c=0$

then, center $\left(\frac{-a}{2}, \frac{-b}{2}\right)$

radius $\left(\sqrt{\left(\frac{-a}{2}\right)^2+\left(\frac{-b}{2}\right)^2-c}\right.$

EX: Find the radius and the center of:

$$x^2+y^2-4x+6y-11=0$$

1 ** center $\left(\frac{--4}{2}, \frac{-6}{2}\right) = (2, -3)$

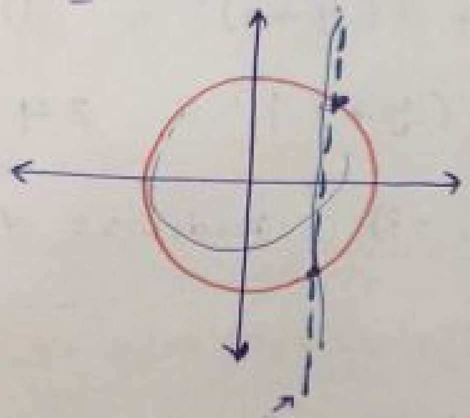
** radius $\sqrt{(2)^2+(-3)^2-(-11)} = \sqrt{24}$

? Is the circle function

Graphically

⇒ By vertically line test

اختبار الخط العمودي.



No By vertical line test

Gebraically (جبرياً)

$$x^2 - y^2 = r^2$$

$$y^2 = x^2 - r^2$$

$$y = \pm\sqrt{x^2 - r^2}$$

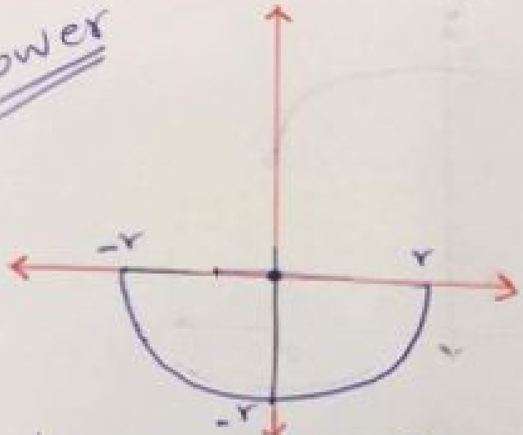
is not function

* تذكر أن الاقتران يجب أن يكون له صورة واحدة لكل عنصر في المجال.

الدائرة ليست اقتران لأن هنالك أعداد تعوض في معادلتها وتعطي قيمتين للمجال.

* lower / upper circle function:

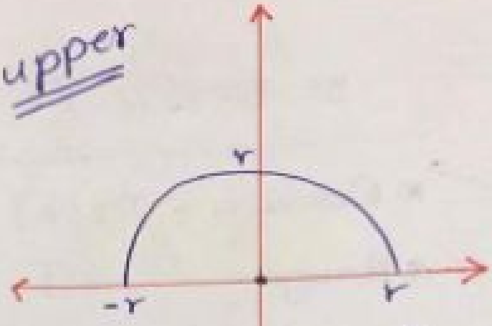
lower



lower circle function

* $y = -\sqrt{r^2 - x^2}$.
* Domain: $[-r, r]$
* Range: $[-r, 0]$

upper



upper circle function

$y = +\sqrt{r^2 - x^2}$

* Domain is: $[-r, r]$
* Range is: $[0, r]$

لاحظ ان ما داخل جذر تربيعي
وحيدا بجد ثابت وهو (r^2).

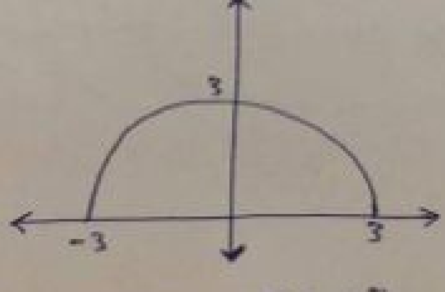
EX: Sketch and find the domain and the Range of the following function:

① $y = +\sqrt{9 - x^2}$.
② $y = -\sqrt{16 - x^2}$.
③ $y = \boxed{5+}\sqrt{4 - x^2}$
④ $y = -3 - \sqrt{1 - x^2}$

By shifting

Sol:

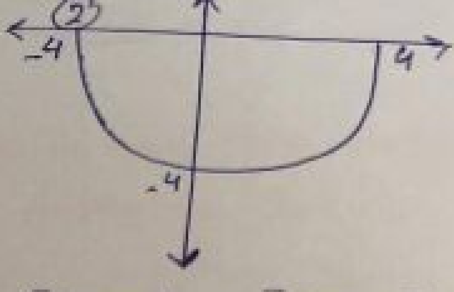
①



Domain = $[-3, 3]$

Range = $[0, 3]$

②

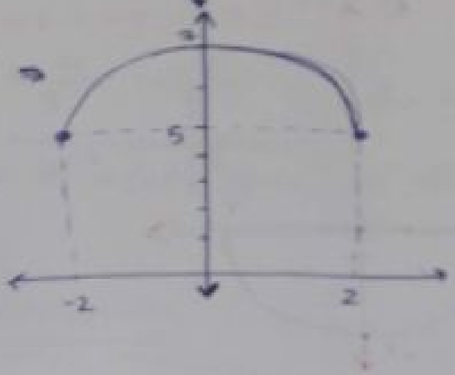


Domain = $[-4, 4]$.

Range = $[-4, 0]$.

③ $\sqrt{4-x^2}$ ⇒

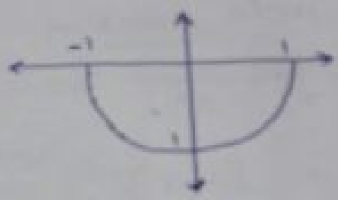
$y = 5 + \sqrt{4-x^2}$ ⇒



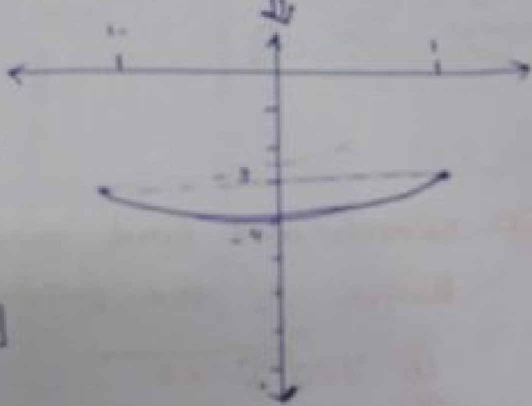
* Domain = [-2 ; 2]
* Range = [5 , 7]

④ $-\sqrt{1-x^2}$

y=



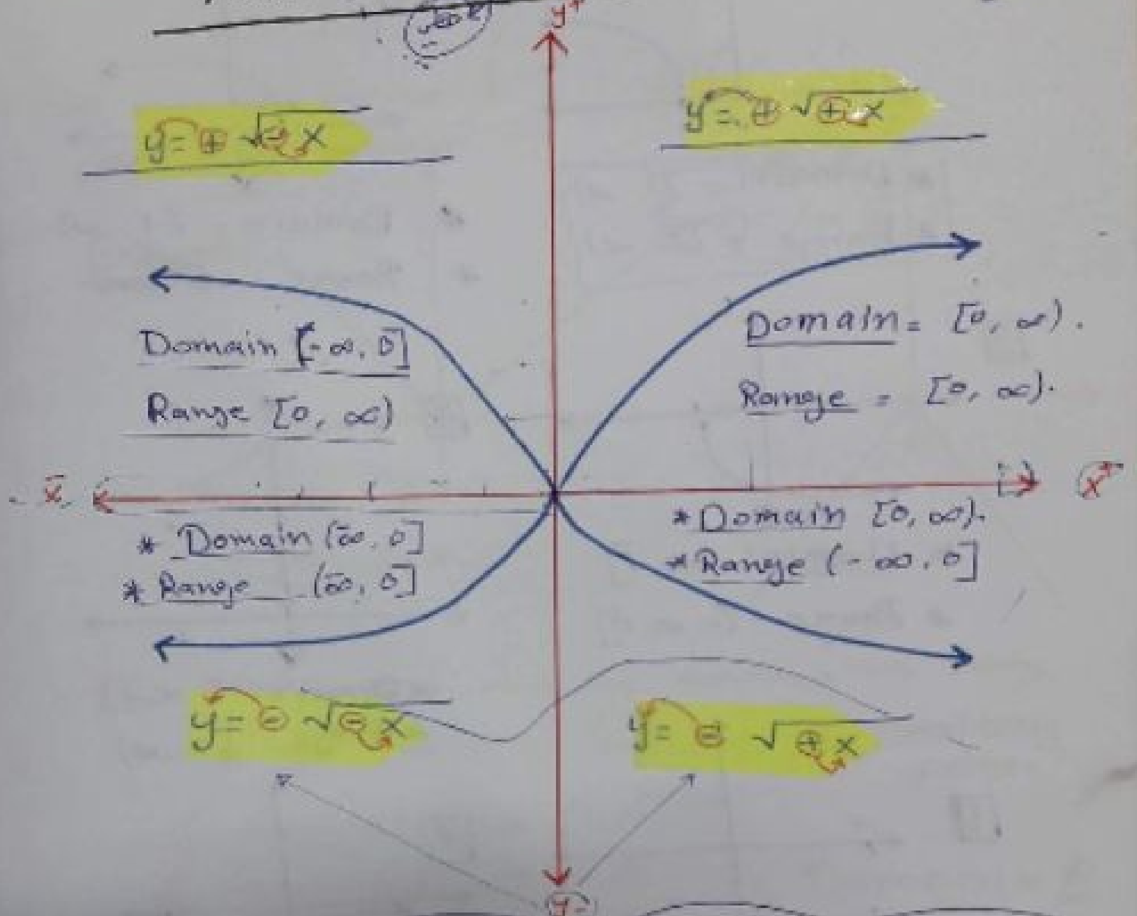
$y = -3 - \sqrt{1-x^2}$



Domain = [-1, 1]
Range = [-4, -3]

(7) The sequare root function of liner express.

$$f(x) = a + \sqrt{bx+c} \quad \text{OR} \quad f(x) = a - \sqrt{bx+c}$$



Graph is apart of $\boxed{C}$ letter toward the x-axis

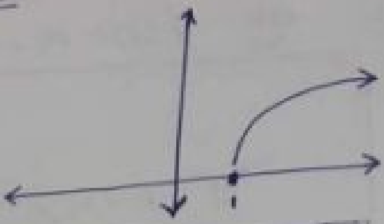
EX: sketch and find the domain and the Range of the following function:

1. $y = \sqrt{x-1}$ → st.pt $\boxed{(1,0)}$
2. $y = \sqrt{x+1}$ → st.pt $(-1,0)$
3. $y = -\sqrt{-x+1}$ → // $(1,0)$
4. $y = 3 + \sqrt{2-x}$ → $(2,3)$
5. $y = -3 - \sqrt{x+4}$. → $(-4,-3)$
6. $y = -2 + \sqrt{6-2x}$ → $(3,-2)$

Solution By starting point

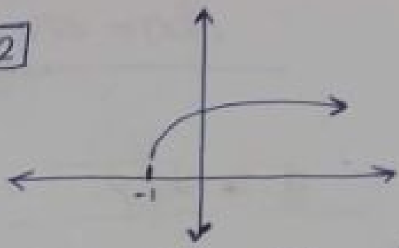
Solution:

[1]



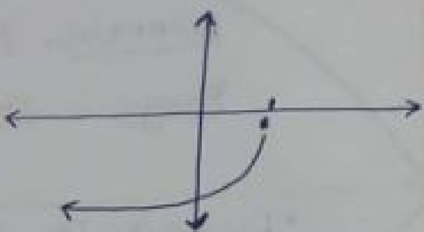
* Domain = [1, ∞)
* Range = [0, ∞)

[2]



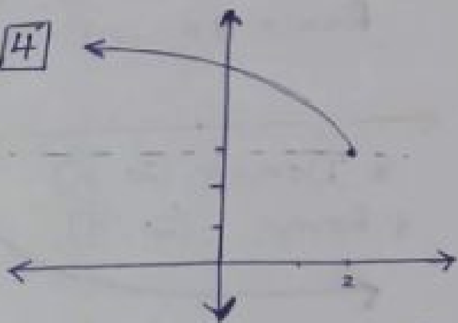
* Domain = [-1, ∞)
* Range = [0, ∞).

[3]



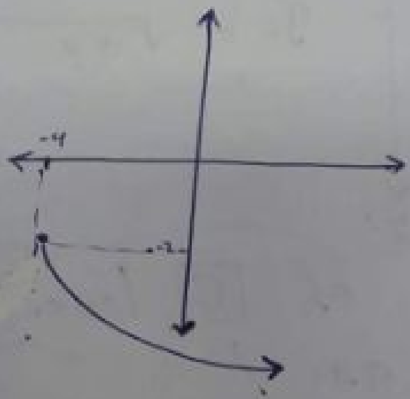
* Domain: (-∞, 1]
* Range: (-∞, 0]

[4]



* Domain: (-∞, 2]
* Range: [3, ∞)

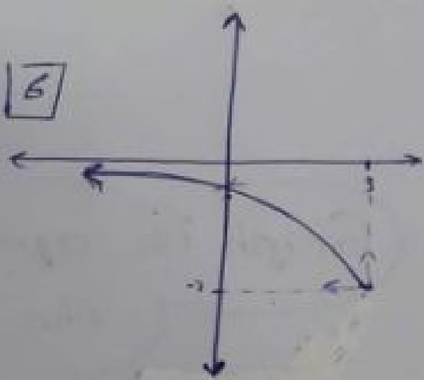
[5]



Domain:

Range:

[6]



* Domain

* Range

8 Absolute value function...

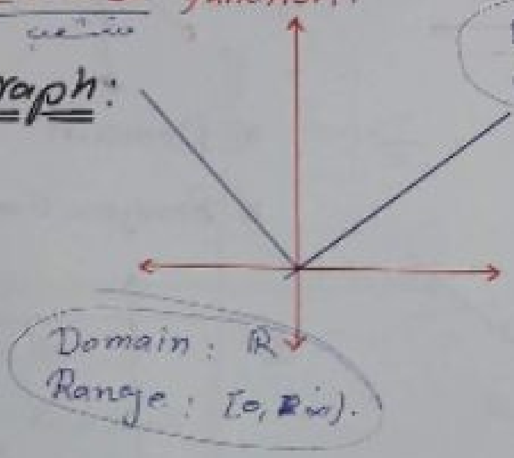
"اقتـران القيمة المطلقة"

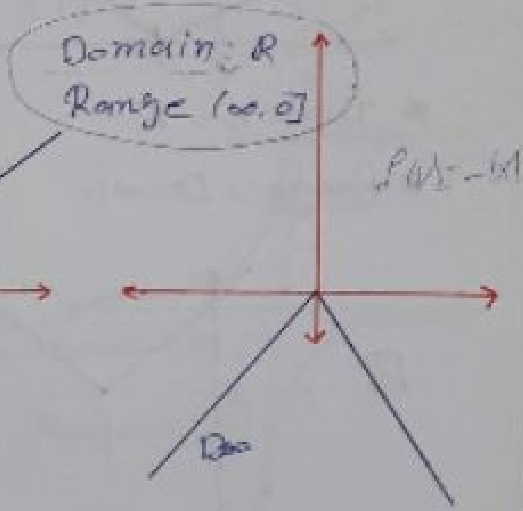
$$f(x) = |x| = \begin{cases} x , & \text{if } x \geq 0 \\ -x , & \text{if } x < 0 \end{cases}$$

هي صيغة اقتران متشعب/متفرع

pice - wise function.
متشعب

Graph:





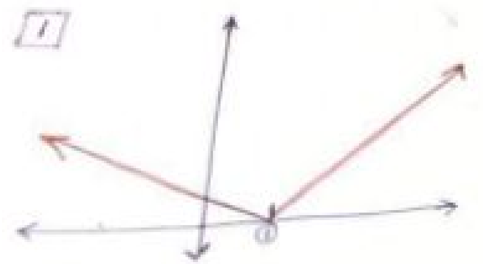
EX: sketch and find the domain and range of the following function: By starting point.

1. $f(x) = |x-1|$. $\rightarrow$ $(1, 0)$
2. $f(x) = -|x-3|$. $\rightarrow$ $(3, 0)$
3. $f(x) = 4 + |2x-4|$ $(2, 4)$
4. $f(x) = -3-|5-x|$. $(5, -3)$
5. $f(x) = |x| + 4$. $(0, 4)$
6. $f(x) = 6 - |x|$. $(0, 6)$

Domain = $\mathbb{R}$

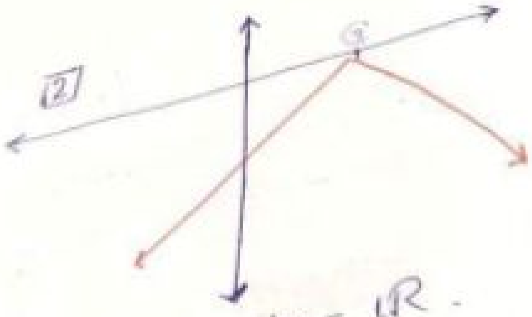
Solution ④ :

1



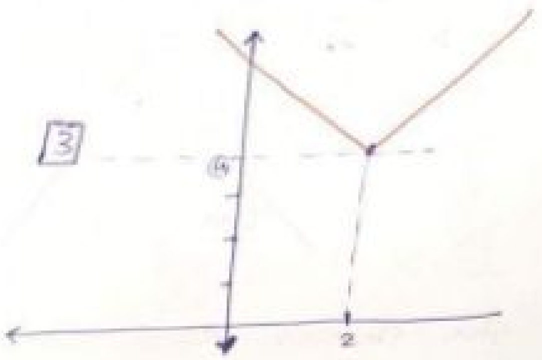
* Domain = $\mathbb{R}$
* Range = $[0, \infty)$.

2



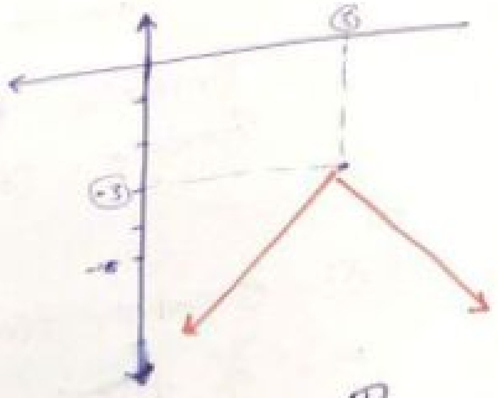
* Domain = $\mathbb{R}$.
* Range = $(-\infty, 0]$.

3



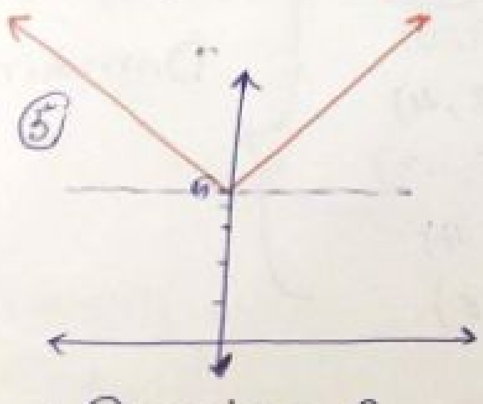
* Domain = $\mathbb{R}$
* Range = $[4, \infty)$.

④

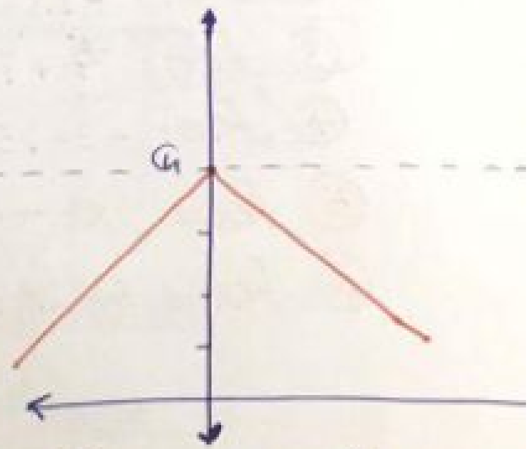


* Domain = $\mathbb{R}$
* Range = $(-\infty, -3]$.

⑤



* Domain = R
* Range = $[4, \infty)$.



Domain = R
Range = $(-\infty, 4]$.

⑨ The rational function of the from:

ملاحظة: معامل (x) في المقام يجب ان يكون موجب.

$$f(x) = \frac{a}{bx+c} \quad \text{OR} \quad f(x) = \frac{-a}{bx+c}$$

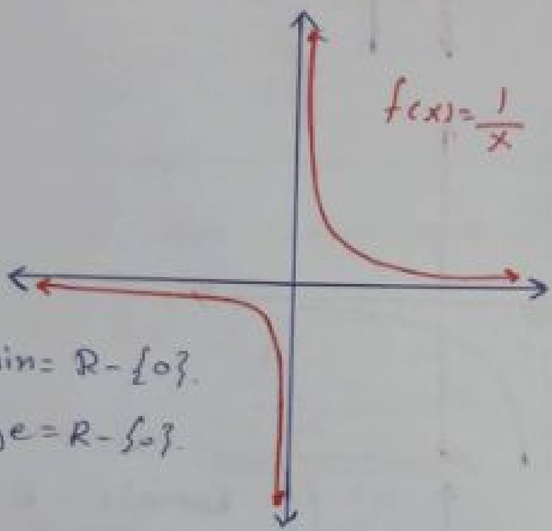
ثابت فقط

متغير في المقام

* Domain = $\mathbb{R} - \{\frac{-c}{b}\}$
* Range = $\mathbb{R} - \{0\}$.

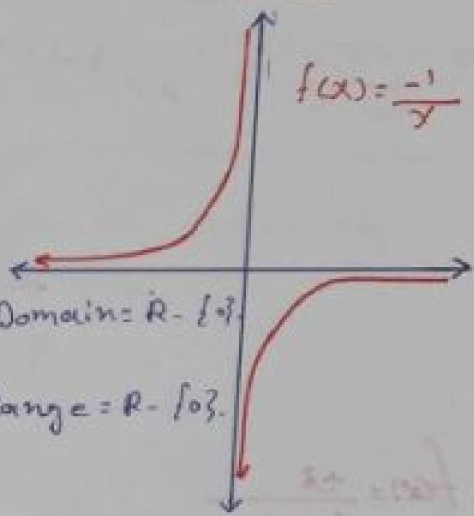
$b > 0$

شرط b ان



Domain = $\mathbb{R} - \{0\}$.

Range = $\mathbb{R} - \{0\}$.



Domain = $\mathbb{R} - \{0\}$.

Range = $\mathbb{R} - \{0\}$.

EX: sketch and find the domain and Range of the following functions:

Solution by vertical line (V.L).

لـ هي نقطة صفر المقام غير متحول عندها

① $f(x) = \frac{1}{x-1}$ ⟶ V.L ($x=1$)

② $f(x) = \frac{-1}{x+4}$ ⟶ V.L ($x=-4$)

③ $f(x) = \frac{3}{4-x}$ ⟶ V.L ($x=4$)

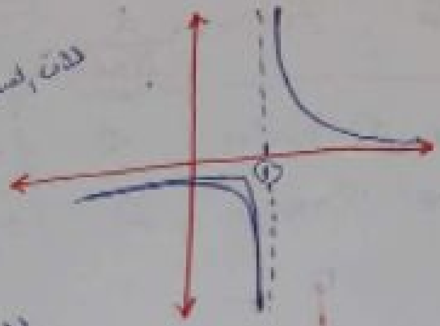
④ $f(x) = \frac{-2}{4-2x}$ ⟶ V.L ($x=2$)

$\frac{-3}{x-9}$

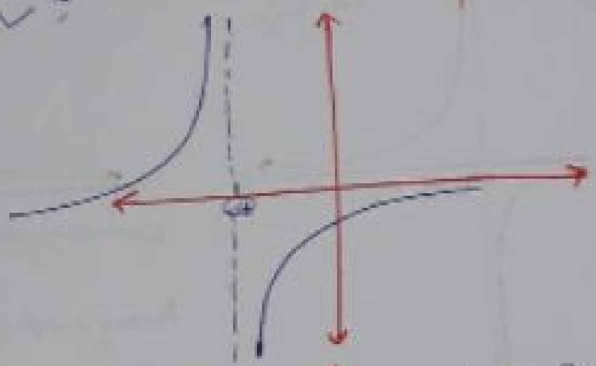
Solution:

①

دالة البسط موجب



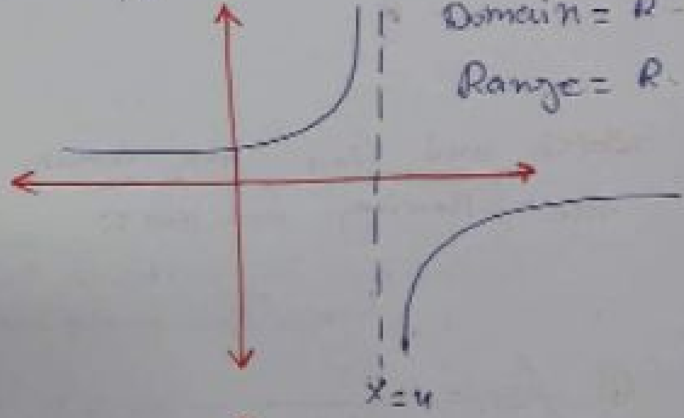
دالة البسط سالب



$f(x) = \frac{+3}{4-2x}$

But:

$f(x) = \frac{-3}{2x-4}$



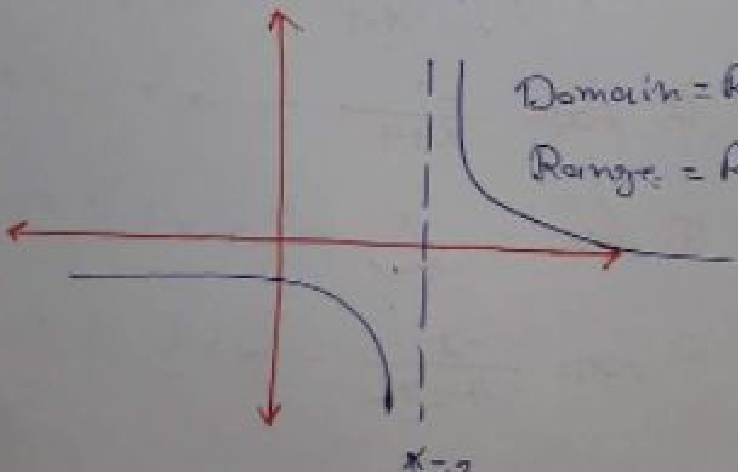
Domain = R - {2}.

Range = R - {0}

$f(x) = \frac{-2}{4-2x}$

But

$f(x) = \frac{2}{2x-4}$



Domain = R - {2}.

Range = R - {0}.

⑩ More functions:

اقترانات أخرى

① $\text{Domain}\{f(x) \overset{\times}{\pm} g(x)\} = \text{Domain}\{f(x)\} \boxed{\cap} \text{Dom}\{g(x)\}$

② $\text{Domain}\{f(x)/g(x)\} = \text{Domain}\, f(x) \cap \text{Domain}\{g(x)\}$
$- \{x : g(x) = 0\}$
اصفار المقام

EX: Find the domain and for each of following.

① $f(x) = \sqrt{(x-1)} + \sqrt{9-x^2}$

② $f(x) = \dfrac{\sqrt{x^2-16}}{x^2-25}$

H.W ③ $f(x) = \dfrac{x}{x-2} - \sqrt{x^2+4x}$

H.W ④ $f(x) = (\sqrt[5]{x^2+4})(\sqrt{x})$.

Solution:-

① $f(x) = \sqrt{(x-1)} + \sqrt{9-x^2}$

$\text{Domain } f(x) = \text{Domain}\{\sqrt{(x-1)}\} \cap \text{Domain}\{\sqrt{9-x^2}\}$

$\sqrt{9-x^2} \downarrow$ upper circle

Domain $[-3, 3]$

$\sqrt{(x-1)} \downarrow$ $x - 1 \geq 0$

sign of $x-1$: $-\infty \quad - \quad 1 \quad + \quad +\infty$

Domain $[1, \infty)$.

$\text{Domain } f(x) = [1, \infty) \cap [-3, 3] = [1, 3]$

② $f(x) = \frac{\sqrt{x^2-16}}{x^2-25}$.

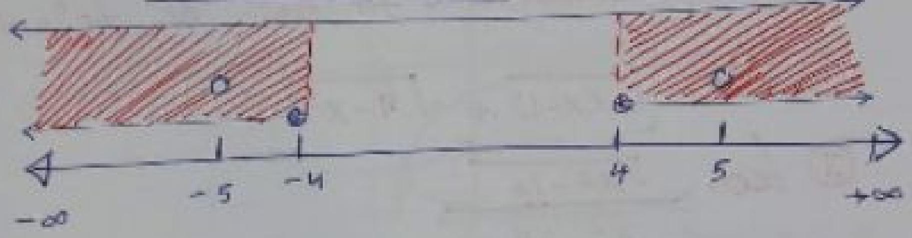
Domain { } = Domain $\{\sqrt{x^2-16}\} \cap$ Domain $\{x^2-25\} - \{x : x^2-25=0\}$

$x^2-16 \geq 0$ $\qquad$ $\mathbb{R}$ (polynomial) $\qquad$ $x = -5, 5$

$x = \pm 4$

sign of x^2-16: $+ \;|_{-4}\; - \;|_{4}\; +$

Domain = $(-\infty,-4] \cup [4,\infty)$.



* Domain $f(x) = (-\infty,-4] - \{5\} \cup [4,\infty) - 5$

or $= (-\infty,-5) \cup (-5,4] \cup [4,5) \cup (5,\infty)$.

or

$= (-\infty,-4] \cup [4,\infty) - \{-5,5\}$.

③ $f(x) = \frac{x}{x-2} - \sqrt{x^2+4x}$.

Domain $\{f x\}$ = Domain $\left\{\frac{x}{x-2}\right\} \cap$ Domain $\{\sqrt{x^2+4x}\}$

$\mathbb{R} - \{2\}$ *

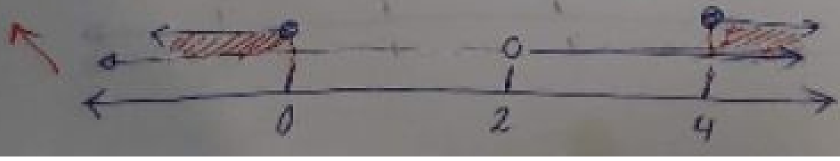
$x^2+4x \geq 0$

$\boxed{x=0,\ x=4}$

$+ \;|_{0}\; - \;|_{4}\; +$

$(-\infty,0] \cup [4,\infty)$.

Domain: $(-\infty,0] \cup [4,\infty)$.



(11) **Composition function.**

تركيب الاقترانات

(ق∘هـ)(س) = ق(هـ(س))

$$(f \circ g)(x) = f(g(x))$$

* Domain of $(f \circ g)(x)$ = Domain $f(g(x)) \cap$ Domain $g(x)$.

* Domain of $(g \circ f)(x)$ = Domain $g(f(x)) \cap$ Domain $f(x)$

الاقتران بعد التركيب — الاقتران الداخلي.

Ex: Given $f(x) = x^2$ and $g(x) = \sqrt{x-1}$

Evalute:

① $(f \circ g)(5)$ ② $(g \circ f)(3)$

Solution

① $f(g(5)) = f(\underbrace{2}_{g(5)}) = (2)^2 = 4$

② $g(f(3)) = g(9) = \sqrt{8}$

Ex: Determine $(f \circ g)(x)$ and $(g \circ f)(x)$ and find the domain of $(f \circ g)(x)$ and $(g \circ f)(x)$.

① $f(x) = x^2$, $g(x) = 1 + x$

② $f(x) = (\sqrt{x} + 1)$, $g(x) = x^2$

③ $f(x) = \dfrac{1}{x+1}$ $g(x) = \sqrt{x} + 1$

④ $f(x) = \dfrac{1}{x-1}$ $g(x) = \dfrac{1}{x-2}$

① → $(f\circ g)_{(x)} = f(g(x)) = f(1+x) = (1+x)^2 = 1+2x+x^2$

→ $(g\circ f)_{(x)} = g(f(x)) = g(x^2) = 1+x^2$

* Domain $(f\circ g)_{(x)}$ = Domain $f(g(x)) \cap$ Domain $g(x)$

Domain $(1+2x+x^2) \cap$ Domain $(1+x)$

$\downarrow \qquad \downarrow$

$\mathbb{R} \qquad \mathbb{R}$

∴ Domain $(f\circ g)_{(x)} = R$.

* Domain $(g\circ f)_{(x)}$ = Domain $g(f(x)) \cap$ Domain $f(x)$

Domain $\{1+x^2\} \cap$ Domain $\{x^2\}$

$\downarrow \qquad \downarrow$

$\mathbb{R} \qquad \mathbb{R}$

∴ Domain $(g\circ f)_{(x)} = \mathbb{R}$.

* ② → $(f\circ g)_{(x)} = f(g(x)) = f(x^2) = \sqrt{x^2}+1 = |x|+1 = x+1$

→ $(g\circ f)_{(x)} = g(f(x)) = g(\sqrt{x}+1) = (\sqrt{x}+1)^2 = x+2\sqrt{x}+1$

* Domain $(f\circ g)_{(x)}$ = Domain $f(g(x)) \cap$ Domain $g(x)$

Domain $\{x+1\} \cap$ Domain $\{x^2\}$

$\downarrow \qquad \downarrow$

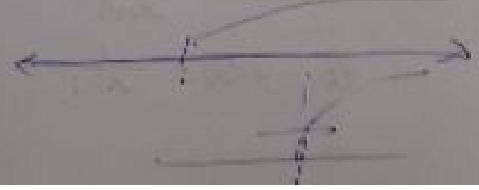
$R \qquad R$

Domain $(f\circ g)_{(x)} = R$.

* ~~Domain g(f(x))=~~

* Domain $(g\circ f)_{(x)}$ = Domain $g(f(x)) \cap$ Domain $f(x)$

Domain $\{x+2\sqrt{x}+1\} \cap$ Domain $\{\sqrt{x}+1\}$



Ex: Find $g(x)$ if:

① $f(x) = x^2$ and $(f \circ g)(x) = (1+x)^2$

② $f(x) = \dfrac{x+1}{x-2}$ and $(g \circ f)(x) = \dfrac{3}{x-2}$ ⇒ مهم جداً عادة ما يتكرر في الامتحان

Remark

- $(\sqrt{x})^2 = x$
- $(\sqrt{x^2}) = |x|$

---

solution: ① $(f \circ g)(x) = f(g(x)) = (1+x)^2$

$$\Rightarrow (g(x))^2 = (1+x)^2$$

$$\Rightarrow \sqrt{(g(x))^2} = \sqrt{(1+x)^2}$$

$$\Rightarrow |g(x)| = |(1+x)|$$

$$\Rightarrow \boxed{g(x) = (1+x)} \text{ or } \boxed{g(x) = -(1+x)}$$

② $(g \circ f)(x) = \dfrac{3}{x-2} \Rightarrow g(f(x)) = \dfrac{3}{x-2}$

$$\Rightarrow g\left(\frac{x+1}{x-2}\right) = \frac{3}{x-2}$$

اولاً نفرض $= y$

$$\Rightarrow g(y) = \frac{3}{\frac{(2y+1)}{(y-1)} - 2}$$

$$g(y) = \frac{3(y-1)}{2y+1-2y+2}$$

$$g(y) = \frac{\cancel{3}(y-1)}{\cancel{3}} = (y-1)$$

Set:

$y = \dfrac{x+1}{x-2}$

$yx - 2y = x + 1$

$yx - x = 2y + 1$

$x(y-1) = (2y+1)$

$x = \dfrac{(2y+1)}{(y-1)}$

الهدف من هذه الخطوة إخراج قيمة x بالنسبة لـ (y).

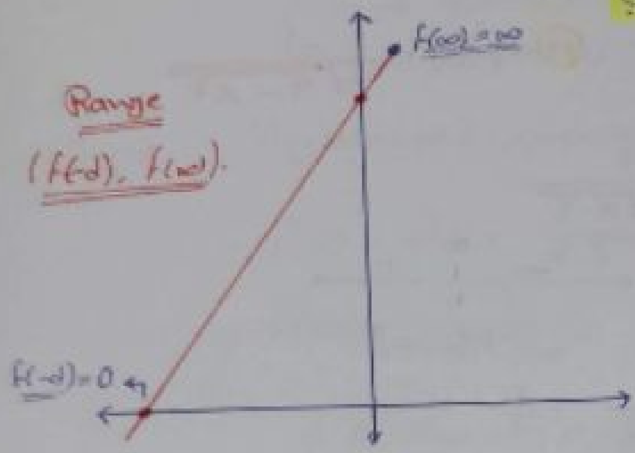
(12) Range of the following forms:

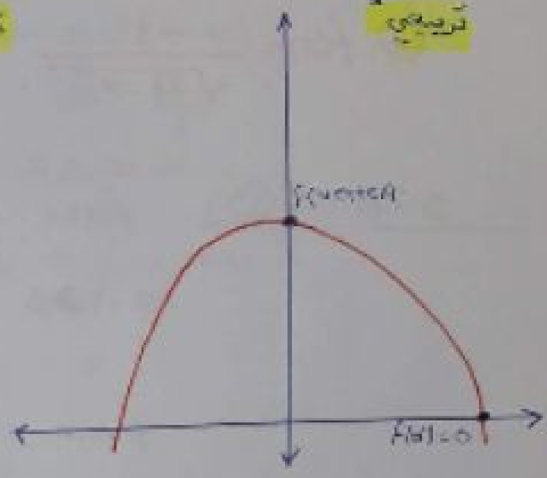
a) $f(x) = \dfrac{c}{\sqrt{ax+b}}$ خطي

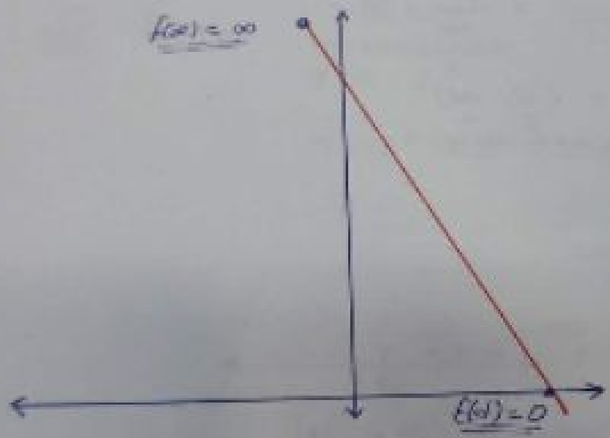
b) $f(x) = \dfrac{c}{\sqrt{ax^2+bx+c}}$ تربيعي

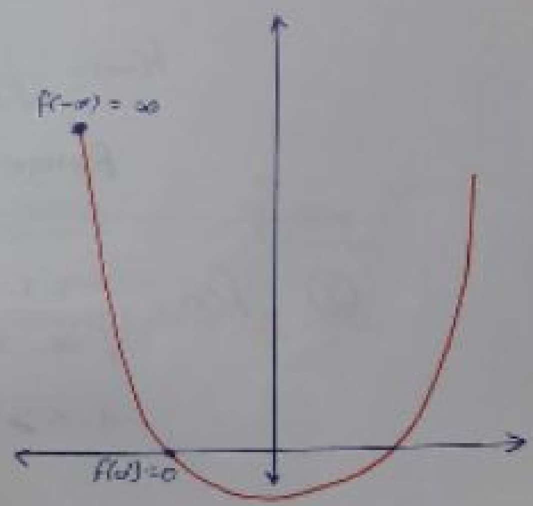
Range

$(f(-d), f(\infty))$.









ملاحظة : دائماً الجذر يرفض القيم السالبة للـ (Range .

Remark:

* $\dfrac{1}{\infty} = 0$
* $\dfrac{1}{0^+} = \infty$
* $\dfrac{1}{0^-} = \infty$

Ex: find the range of:

① $f(x) = \dfrac{1}{\sqrt{x-1}}$ ③ $f(x) = \dfrac{1}{\sqrt{x^2 - 4x}}$

② $f(x) = \dfrac{-3}{\sqrt{4-x}}$ ④ $f(x) = \dfrac{-1}{\sqrt{9 - x^2}}$

Solution ①: $f(x) = \dfrac{1}{\sqrt{x-1}}$

$x - 1 \geq 0$: $\longleftarrow \; - \; \underset{1}{|} \; + \longrightarrow$

$(1, \infty)$

$f(1) = \dfrac{1}{\sqrt{1-1}} = \dfrac{1}{0} = \infty$

$f(\infty) = \dfrac{1}{\sqrt{\infty - 1}} = \dfrac{1}{\infty} = 0$

Range $= (0, \infty)$.

② $f(x) = \dfrac{-3}{\sqrt{4-x}}$

$4 - x \geq 0$: $\longleftarrow + + + \underset{4}{|} - - - \longrightarrow$

Domain $(\infty, 4)$

• $f(\infty) = \dfrac{-3}{\sqrt{4-\infty}} = \dfrac{-3}{\infty} = 0$

• $f(4) = \dfrac{-3}{\sqrt{4-4}} = \dfrac{-3}{0^+} = -\infty$

Range $(-\infty, 0)$.

③ $f(x) = \dfrac{1}{\sqrt{x^2 - 4x}}$

$x^2 - 4x \geq 0$ + 0 ⁄⁄⁄ 0 + (0, 4)

Igrore
تجاهل

→ Domain $= (-\infty, 0) \cup (4, \infty)$.

→ Vertex at $x = \frac{-b}{2a} = \boxed{2}$

↳ خارج الـ (Domain) لذلك نتجاهلها " Ignor.

- $f(4) = f(0) = \dfrac{1}{\sqrt{0}} = \dfrac{1}{0} = \infty$

- $f(\infty) = \dfrac{1}{\sqrt{\infty^2 - 4\infty}} = \dfrac{1}{\infty} = 0$

Range $= (0, \infty)$.

④ $f(x) = \dfrac{-1}{\sqrt{9 - x^2}}$

$9 - x^2 \geq 0$: Igrore −3 0 + 0 3 Igrore.

Domain $= (-3, 3)$.

vertex at $x = \frac{-b}{2a} = 0$

$f(-3) = f(3) = -\infty$

$f(0) = \frac{1}{3}$

Range $= (-\infty, \frac{1}{3}]$.

لأنها ضمن الفترة

* Limits — النهايات

⑭ Limits as $x \to \infty$ or $x \to -\infty$;

Then: ① $\lim\limits_{x \to \infty} \frac{1}{x} = \frac{1}{\infty} = 0.$

② $\lim\limits_{x \to -\infty} \frac{1}{x} = \frac{1}{-\infty} = 0.$

Ex: Find the following limits:

① $\lim\limits_{x \to \infty} \frac{3x^2 - 5x + 1}{x^2 + x + 2}$

② $\lim\limits_{x \to \infty} \frac{x^2 + 1}{2x^3 - x}$

③ $\lim\limits_{x \to \infty} \frac{x^3 - 8}{x^2 - 4}$

④ $\lim\limits_{x \to \infty} \frac{2x + 1}{\sqrt{x^2 + 1}}$

⑤ $\lim\limits_{x \to \infty} \frac{2x + 3}{\sqrt{9x^2 - 1}}$

⑥ $\lim\limits_{x \to \infty} \frac{|x| + |x - 2|}{x}$

⑦ $\lim\limits_{x \to -\infty} \frac{|x| + |x - 2|}{x}$

خطوات كل: ١- اخراج أعلى درجة في أوس في البسط، أعلى درجة أوس في المقام.

٢- نعوض (∞) في المعادلة.

① $\lim\limits_{x \to \infty} \frac{\cancel{x^2}(3 - \frac{5}{x} + \frac{1}{x^2})}{\cancel{x^2}(1 + \frac{1}{x} + \frac{2}{x^2})} = \frac{3 - \frac{5}{\infty} + \frac{1}{\infty^2}}{1 + \frac{1}{\infty} + \frac{2}{\infty^2}} = \frac{3 - 0 + 0}{1 + 0 + 0} = 3$

② $\lim\limits_{x \to \infty} \frac{\cancel{x^2}(1 + \frac{1}{x^2})}{\cancel{x^3}(2 - \frac{1}{x^2})} = \lim\limits_{x \to \infty} \frac{1 + \frac{1}{x^2}}{2 - \frac{1}{x^2}} = \frac{1 + \frac{1}{\infty}}{2 - \frac{1}{\infty}} = \frac{1}{2}$

③ $\lim\limits_{x \to \infty} \frac{\cancel{x^3}^{\,x}(1 - \frac{8}{x^3})}{\cancel{x^2}(1 - \frac{4}{x^2})} = \frac{\infty(1 - \frac{8}{\infty^3})}{(1 - \frac{4}{\infty^2})} = \stackrel{+}{\infty}$

Remark:

$|x| = x$ when $x \to \infty$.

$|x| = -x$ when $x \to -\infty$.

$|-x| = -x$ when $x \to -\infty$.

$|-x| = x$ when $x \to \infty$.

(4) $\lim\limits_{x \to \infty} \dfrac{x(2+\frac{1}{x})}{(\sqrt{x^2}\sqrt{1+\frac{1}{x^2}})} \Rightarrow \lim\limits_{x \to \infty} \dfrac{x(2+\frac{1}{x})}{|x|\sqrt{1+\frac{1}{x^2}}} = \lim\limits_{x \to \infty} \dfrac{\cancel{x}(2+\frac{1}{x})}{\cancel{x}(1+\frac{1}{x^2})}$

$= 2$

(5) $\lim\limits_{x \to -\infty} \dfrac{2x+3}{\sqrt{9x^2-1}} \Rightarrow \lim\limits_{x \to -\infty} \dfrac{\cancel{x}(2+\frac{3}{x})}{-\cancel{x}\sqrt{9-\frac{1}{x^2}}} = \dfrac{2}{-3}$

(6) $\lim\limits_{x \to \infty} \dfrac{|x|+|x-2|}{x} \Rightarrow \lim\limits_{x \to \infty} \dfrac{|x+x-2|}{x} = \lim\limits_{x \to \infty} \dfrac{|2x-2|}{x}$

$\lim\limits_{x \to \infty} \dfrac{\cancel{x}(2-\frac{2}{x})}{\cancel{x}} = 2$

(7) $\lim\limits_{x \to -\infty} \dfrac{|x|+|x-2|}{x} = \lim\limits_{x \to -\infty} \dfrac{|2x-2|}{x} = \lim\limits_{x \to -\infty} \dfrac{2-2x}{x}$

$\lim\limits_{x \to -\infty} \dfrac{\cancel{x}(\frac{2}{x}-2)}{\cancel{x}} = -2$

* اختصاراً للجهد، الوقت
دائماً جواب هذا النوع من السؤالات
يكون: $\dfrac{\text{معامل أعلى أس في البسط}}{\text{معامل أعلى أس في المقام}}$.

مثلاً في السؤال الأول:

$\lim\limits_{x \to \infty} \dfrac{3x^2-5x+1}{x^2+x+2}$

* معامل أعلى أس في البسط = 3
* معامل أعلى أس في المقام = 1

الجواب = $\dfrac{3}{1}$

الامتحان الاول
السبت ٢٩/١٠/٢٠١٦
من الساعة 9.15 ← 10.15
القاعة SB13
رقم التسلسل 45

15 Limits As $\to a$

Ex: Evalute the following:

1. $\lim\limits_{x\to 2} (3x^2+7x-1)$ .25 ✓

4. $\lim\limits_{x\to 2^-} \dfrac{x^2-x-2}{|x-2|}$ $\quad \frac{0}{0}$

2. $\lim\limits_{x\to 5} \dfrac{x^2-25}{\sqrt{x^2+1}}$ $\quad \frac{0}{\sqrt{26}} = 0$

5. $\lim\limits_{x\to 1^-} \dfrac{(x^3-1)}{(x^2-1)}$ $\quad \frac{0}{0}$

3. $\lim\limits_{x\to 0} \dfrac{\sqrt{4+x}-2}{x^2+4x}$ $\quad \frac{0}{0}$

H.W 6. $\lim\limits_{x\to 0} \dfrac{\sqrt{x+1}-1}{\sqrt{x+4}-2}$ $\quad \frac{0}{0}$

Solution 3: $\lim\limits_{x\to 0} \dfrac{\sqrt{4+x}-2}{x(x-4)} * \dfrac{\sqrt{4+x}+2}{\sqrt{4+x}+2} =$

$\lim\limits_{x\to 0} \dfrac{\cancel{4}+\cancel{x}-\cancel{4}}{\cancel{x}(x-4) * (\sqrt{4+x}+2)} = \dfrac{1}{16}$

4. $\lim\limits_{x\to 2^-} \dfrac{x^2-x-2}{|x-2|} = \lim\limits_{x\to 2^-} \dfrac{\overset{-1}{\cancel{(x-2)}}(x+1)}{\cancel{(2-x)}} = -3$

5. $\lim\limits_{x\to 1^-} \dfrac{(x^3-1)}{(x^2-1)} = \lim\limits_{x\to 1^-} \dfrac{\cancel{(x-1)}(x^2+x+1)}{\cancel{(x-1)}(x+1)} = \dfrac{3}{2}$

6. $\lim\limits_{x\to 0} \dfrac{\sqrt{x+1}-1}{\sqrt{x+4}-2} * \dfrac{\sqrt{x+1}+1}{\sqrt{x+1}+1} * \dfrac{\sqrt{x+4}+2}{\sqrt{x+4}+2}$

$\Rightarrow \lim\limits_{x\to 0} \dfrac{\cancel{x}+\cancel{1}-\cancel{1}}{\cancel{x}+4-4} * \dfrac{(\sqrt{x+4}+2)}{(\sqrt{x+1}+1)} = \dfrac{4}{2} = 2$

Ex: Find the following limits:

① $\lim\limits_{x \to 2^-} \dfrac{1}{x-2}$ $\quad \frac{1}{0}$

③ $\lim\limits_{x \to -2^-} \dfrac{1}{x+2}$ $\quad \frac{1}{0}$

② $\lim\limits_{x \to 1^+} \dfrac{1}{x-1}$ $\quad \frac{1}{0}$

* ④ $\lim\limits_{x \to -2^-} \dfrac{1}{x-2}$ $\quad \frac{1}{-4}$

Solution

Remark:

$a^+ - a = 0^+$

$a^- - a = 0^-$

$-a^+ - a = 0^+$

$-a^- - a = 0^-$

Ex:

$2^+ - 2 = 0^+$

$2^- - 2 = 0^-$

$-2^+ + 2 = 0^+$

$-2^- + 2 = 0^-$

① $\lim\limits_{x \to 2^-} \dfrac{1}{x-2} = \dfrac{1}{2^- - 2} = \dfrac{1}{0^-} = -\infty$

② $\lim\limits_{x \to 1^+} \dfrac{1}{x-1} = \dfrac{1}{1^+ - 1} = \dfrac{1}{0^+} = +\infty$

③ $\lim\limits_{x \to -2^-} \dfrac{1}{x+2} = \dfrac{1}{-2^- + 2} = \dfrac{1}{0^-} = -\infty$

④ $\lim\limits_{x \to -2^-} \dfrac{1}{x-2} = \dfrac{1}{-2^- - 2} = \dfrac{1}{-4}$

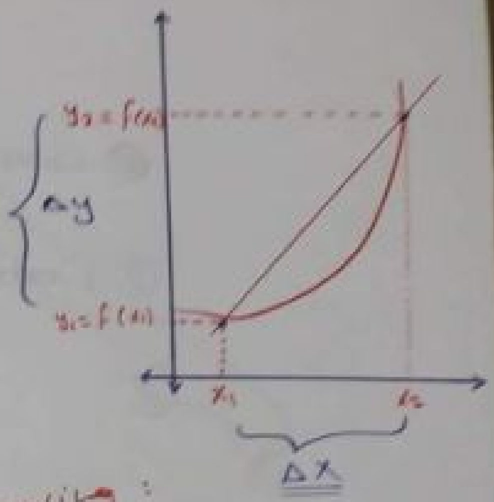
⑰ Average rate of change

معدل التغير

Average rate of change $= \frac{\Delta y}{\Delta x}$.

$$\frac{\Delta y}{\Delta y} = \frac{f(x)_2 - f(x_1)}{\Delta x}$$

$$\frac{\Delta y}{\Delta x} = \frac{f(x_1+\Delta x) - f(x_1)}{\Delta x}$$



Ex: Find the avarge for the following:

① $f(x) = 7-3x$ from $x=2$, $\Delta x = 1$

② $f(x) = \frac{3}{x}$ from x to $x+\Delta x$

Solution: Av. rate of charge $= \frac{\Delta y}{\Delta x} = \frac{f(x_2) - f(x_1)}{\Delta x}$

$$\frac{\Delta y}{\Delta x} = \frac{f(3) - f(2)}{1} = -2 - 1 = -3$$

أو متوسط التغير للاقترانات الخطية يساوي معامل (x).

② Av. rate of charge $= \frac{\Delta y}{\Delta x} = \frac{f(x_2) - f(x_1)}{\Delta x}$

$$= \frac{f(x+\Delta x) - f(x)}{\Delta x} = \frac{\frac{3}{x+\Delta x} - \frac{3}{x}}{\Delta x}$$

$$= \frac{\frac{3 - 3(x+\Delta x)}{x(x+\Delta x)}}{\Delta x}$$

$$= \frac{-3}{x^2 + (\Delta x)(x)}$$

انتهيت مادة الاشتقاق الأول ...

Academic
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Healing

Batch
16

شفاء .